

KARLSTAD SOL, Sweden

WMO: 024150

Lat: 59.3592N

Lon: 13.4719E

Elev: 151

StdP: 14.62

Time Zone: 1.00 (EUC)

Period: 98-11

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-0.1	5.2	-3.6	4.6	0.1	1.6	6.0	6.9	23.8	33.6	21.5	32.8	4.8	350	0.623

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.5	79.1	62.8	76.6	62.5	73.2	61.0	65.8	74.1	64.3	71.9	63.0	70.0	8.5	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.9	86.4	67.9	61.2	81.3	66.8	60.4	78.8	66.1	30.7	73.9	29.5	72.2	28.5	70.0	72.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.2	17.9	15.9	DB	-2.0	84.6	8.2	3.3	-7.9	87.0	-12.7	89.0	-17.3	90.8	-23.3	93.3	(4)
(5)			WB	-3.5	68.7	7.7	1.1	-9.0	69.5	-13.5	70.2	-17.8	70.8	-23.4	71.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.2	27.0	26.2	31.9	42.5	50.3	57.6	62.4	60.2	53.2	43.1	35.2	27.8
	DBStd	14.70	9.69	8.54	7.28	5.65	5.68	4.95	4.69	4.80	5.26	6.62	7.55	10.83
	HDD50	3631	713	665	562	235	63	2	0	1	30	228	443	689
	HDD65	8002	1178	1085	1027	675	455	230	111	159	355	680	893	1154
	CDD50	1153	0	0	0	9	74	230	384	319	125	12	0	0
	CDD65	49	0	0	0	0	0	7	30	12	0	0	0	0
	CDH74	518	0	0	0	0	6	126	286	100	0	0	0	0
	CDH80	65	0	0	0	0	0	18	44	3	0	0	0	0
Wind	WSAvg	7.6	8.3	7.6	7.5	7.4	7.8	7.4	7.0	6.9	7.5	7.5	8.1	7.7
Precipitation	PrecAvg	25.2	1.9	1.3	1.4	1.6	1.8	2.2	2.4	3.0	2.6	2.8	2.8	1.9
	PrecMax	34.3	5.2	3.7	3.8	4.3	4.2	7.3	7.3	6.1	5.4	6.9	6.4	3.5
	PrecMin	17.8	0.1	0.0	0.1	0.0	0.4	0.3	0.2	0.0	0.6	0.6	0.6	0.2
	PrecStd	4.4	1.1	0.8	0.8	0.9	1.0	1.5	1.3	1.6	1.2	1.5	1.4	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.8	46.0	55.0	67.7	73.8	82.5	84.4	80.5	72.0	59.2	52.2	47.9
		MCWB	43.5	42.2	44.4	54.1	55.9	62.6	65.3	63.7	61.5	54.0	49.1	47.6
	2%	DB	43.0	42.6	49.6	62.2	70.2	77.3	80.5	77.1	68.1	56.9	50.2	46.0
		MCWB	41.0	39.6	42.1	50.6	55.1	60.9	63.8	63.4	58.6	53.3	48.1	45.3
	5%	DB	41.0	39.6	44.7	57.6	66.3	73.1	77.1	73.8	64.7	55.0	48.2	44.3
		MCWB	39.1	37.4	38.9	48.0	53.0	59.1	63.1	62.7	57.0	52.0	46.7	43.3
	10%	DB	39.0	37.6	41.3	53.9	62.5	69.4	73.5	70.1	62.5	52.2	46.2	41.4
		MCWB	37.7	35.8	37.6	45.3	51.4	57.7	62.0	61.4	56.7	49.8	45.1	40.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.1	42.5	44.8	55.4	59.9	65.1	68.6	68.1	63.2	56.3	50.8	46.9
		MCDB	45.4	43.4	51.8	61.8	66.9	75.2	78.2	74.5	69.4	58.0	52.1	47.1
	2%	WB	41.4	39.8	42.4	50.9	56.9	62.9	66.7	65.5	60.5	54.2	48.6	44.7
		MCDB	42.3	41.2	47.4	59.3	66.4	73.1	76.0	72.3	65.7	56.2	49.5	45.0
	5%	WB	39.1	37.7	40.3	49.2	54.8	60.7	65.1	63.9	58.9	52.4	46.5	42.6
		MCDB	39.9	39.2	43.2	56.8	63.5	70.4	73.4	70.3	63.3	54.5	47.6	43.2
	10%	WB	37.2	35.8	38.2	46.8	52.7	59.0	63.5	62.5	57.2	50.9	44.7	40.4
		MCDB	38.3	37.5	40.3	52.1	61.0	67.4	70.9	68.5	61.5	52.8	45.4	41.2
Mean Daily Temperature Range	5% DB	MDBR	8.9	10.5	13.6	17.4	18.0	18.2	16.5	16.2	15.5	11.8	8.6	9.3
		MCDBR	9.6	10.7	18.7	23.8	25.3	25.3	23.1	21.2	18.8	12.3	10.4	8.0
		MCWBR	9.3	9.1	12.8	13.9	12.7	11.8	10.4	10.5	11.4	9.1	9.8	8.4
	5% WB	MCDBR	8.8	9.7	15.6	21.4	21.1	21.4	19.5	17.5	16.0	11.0	10.4	8.5
		MCWBR	8.8	8.9	11.3	13.1	11.9	11.2	10.0	10.0	10.5	8.9	10.1	8.8
Clear-Sky Solar Irradiance	taub	0.263	0.304	0.328	0.366	0.363	0.354	0.375	0.367	0.338	0.327	0.288	0.243	
	taud	2.211	2.298	2.388	2.362	2.432	2.481	2.440	2.456	2.523	2.467	2.301	2.135	
	Ebn at Noon	173	226	258	266	275	279	270	263	254	218	162	134	
	Edh at Noon	14	21	27	32	32	31	32	29	23	18	13	10	
All-Sky Solar Radiation	RadAvg	97	279	771	1214	1583	1823	1629	1279	855	379	127	66	
	RadStd	18	54	78	160	151	108	203	128	102	70	30	7	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (63 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page